Structural certificates for the unitary degree-three groups over $\mathbf{F}_4$

Private L054 finite application

Theorem 1. *Relative to the acquired numerical and named-model interfaces, all 234 rows of the three current companion tables have their displayed structural descriptions: 34 rows for $U_3(4)$, 80 for $U_3(4) : 2$, and 120 for $U_3(4) : 2{\cdot}2$. Each label asserts the existence of its stated decomposition, without claiming that the label specifies a unique group.*

Exact source and actual embeddings. The current companion PDFs are attributed to Thomas Connor and Dimitri Leemans. All ten PDF pages were directly viewed, including the final footer-only page. Source powers, parentheses and punctuation are retained: 5^2 is elementary abelian, $4^2 = C_4^2$, and the top expression is $(U_3(4) : 2) \cdot 2$, with a nonsplit final extension. The 2015 article's default left association also applies to the successive dots in $2 \times 4 \cdot 2 \cdot 2 \cdot 2$. The source-row bijections and complete numerical inventories are acquired from the accepted robust reweighting and normal-restriction applications. Both robust input arrays contain only literal permutation groups; the middle table's complete state uses the accepted explicit-generator grammar. No proper subgroup is reconstructed from a displayed constructor name.

The original successful `PSU(3,4)` execution, its exact driver, clean receipt and every recorded input/output pin are acquired. Its actual 273-point generators equal the current top generators. GAP Reference Manual 50.2-14 supplies the named projective special unitary constructor interface. This is the one new named input; a matching order alone does not identify it. The actual two larger groups and all their subgroup embeddings come from the acquired numerical model certificates, with no new automorphism census.

Lemma 2 (Faithful invariant-subset restriction)**.** *Let a finite permutation group G, its order n, and actual subgroups $K_i \leq G$ be acquired. Suppose a literal subset Ω is invariant under every ordered generator of G. Restriction to Ω defines a homomorphism. If its generated image has order n, the homomorphism is injective. Restricting the actual generators of every K_i therefore gives their exact images under this same isomorphism.*

Proof. Each generator permutes the finite subset, so all words preserve it. Restriction commutes with right permutation multiplication, proving the homomorphism assertion. Its kernel has order $|G|/|\operatorname{im}| = 1$. The subgroup assertion follows by applying the same map to its generators. Thus order comparison is used after constructing a specific homomorphism. □

The three literal transports. Each acquired action has an invariant subset of 65 of its 273 points. The independent Python reader checks the complete original generator and subgroup arrays against the acquired files, all permutation domains and all 35,750 generator-point equations. Literal closures of the three image groups have orders 62,400, 124,800 and 249,600, exactly the acquired orders. Consequently all 234 images retain their original embeddings. This replaces neither an ambient subgroup classification nor a named-group identification.

Structural certificates and their proof. The registry acquires 81 earlier accepted models and the named unitary input just described. There are 38 new models: 15 split extensions, 21 nonsplit extensions and two direct products. Each factor and occurrence map is an explicit bijection fixing identity and preserving every right generator edge. The acquired MC.3 word-induction theorem proves these maps are isomorphisms. Normality is checked by conjugating subgroup generators by ambient generators. For a split extension, an actual complement has trivial intersection with the normal factor and the required product of orders. Direct products additionally have commuting factors; nontrivial semidirect actions have explicit noncommuting generator pairs.

For a prime-order quotient, a complement exists exactly when some outside element has that prime order. The reader checks all outside elements, giving 126,604 tests, including every one of the 124,800 elements outside the top index-two normal subgroup. Two other quotients are S_3: their actual normal forms and relations are checked first. The acquired S_3-lift criterion then tests all 425,984 pairs of lifts of its standard generators. Absence of a pair satisfying the presentation proves nonsplitting; failure of a candidate-search routine is not a premise. These criteria and the earlier atom definitions are acquired unchanged. In total, 289 maps and 2,477,559 generator edges prove the displayed structures and join them to the exact source rows.

Finite evidence and limits. The sole bounded witness-discovery pass completed all 234 rows in 20.84 seconds. GAP supplied candidate decompositions and maps only. Replay uses literal permutations; it repeats neither discovery nor any subgroup census. Ordinary and optimized Python must return identical reports. Twenty focused damages cover source rows, dots and powers, original embeddings, named top generators, orbit invariance, actual restriction equations and malformed source syntax. A literal C_2^2 action with an invariant 65-point subset losing one factor is rejected by the image-order guard; a faithful diagonal C_2 control passes. C_4 and C_2^2 distinguish nonsplit and split index-two extensions. Earlier complete transport and extension damage controls are acquired separately.

The first successful control harness used a temporary file in its working directory. The successor moves only that scratch file to a temporary directory, allowing immutable replay; the prior receipt and exact code remain preserved. No mathematical code or certificate changed. This application claims 234 current-table structure joins, with explicit retained classification and software inputs. It claims no historical PDF byte identity, canonical uniqueness of extension labels, whole-source admission, new question resolution, formalization or measurement. Authorship is `1044_resume`; root directed reuse and conducts the distinct review.